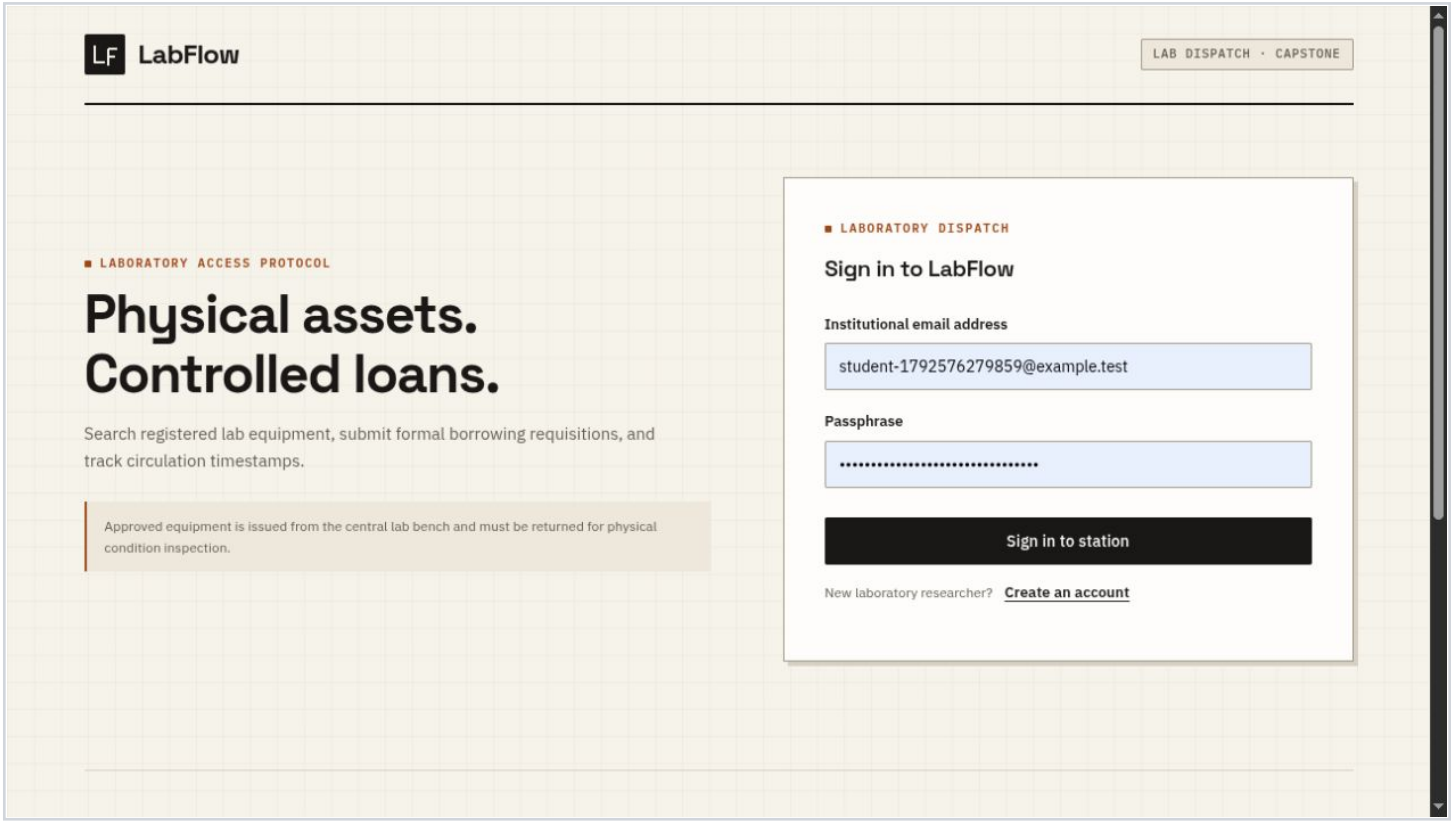
1. Sign-In & Laboratory Access Gate

FEATURE F1 — STUDENT & ADMINISTRATOR AUTHENTICATION

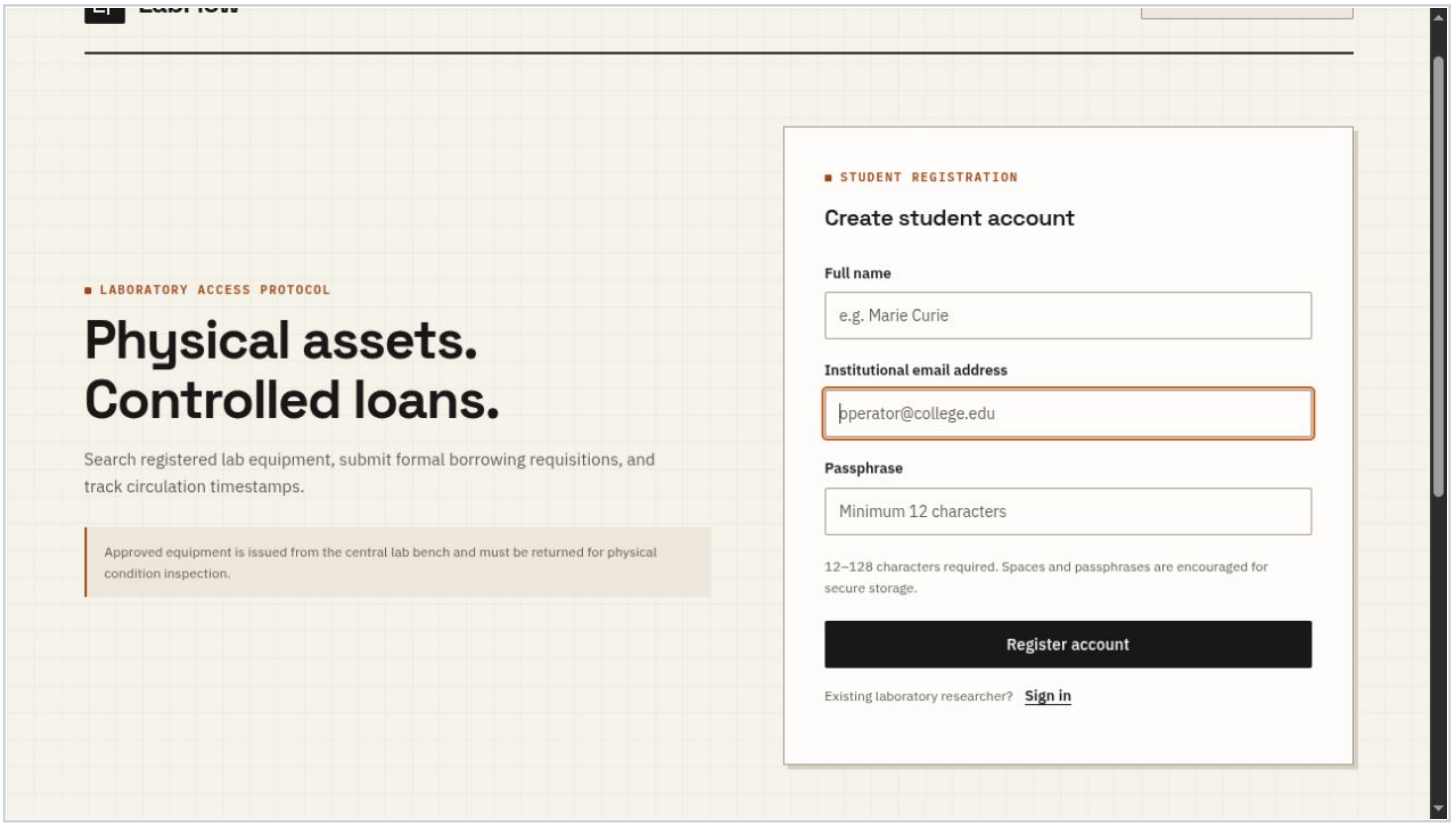
Secure sign-in interface supporting institutional credentials, active session restoration, CSRF tokens, and rate-limited authentication attempts.



2. Student Account Registration

FEATURE F1 — STUDENT SELF-REGISTRATION

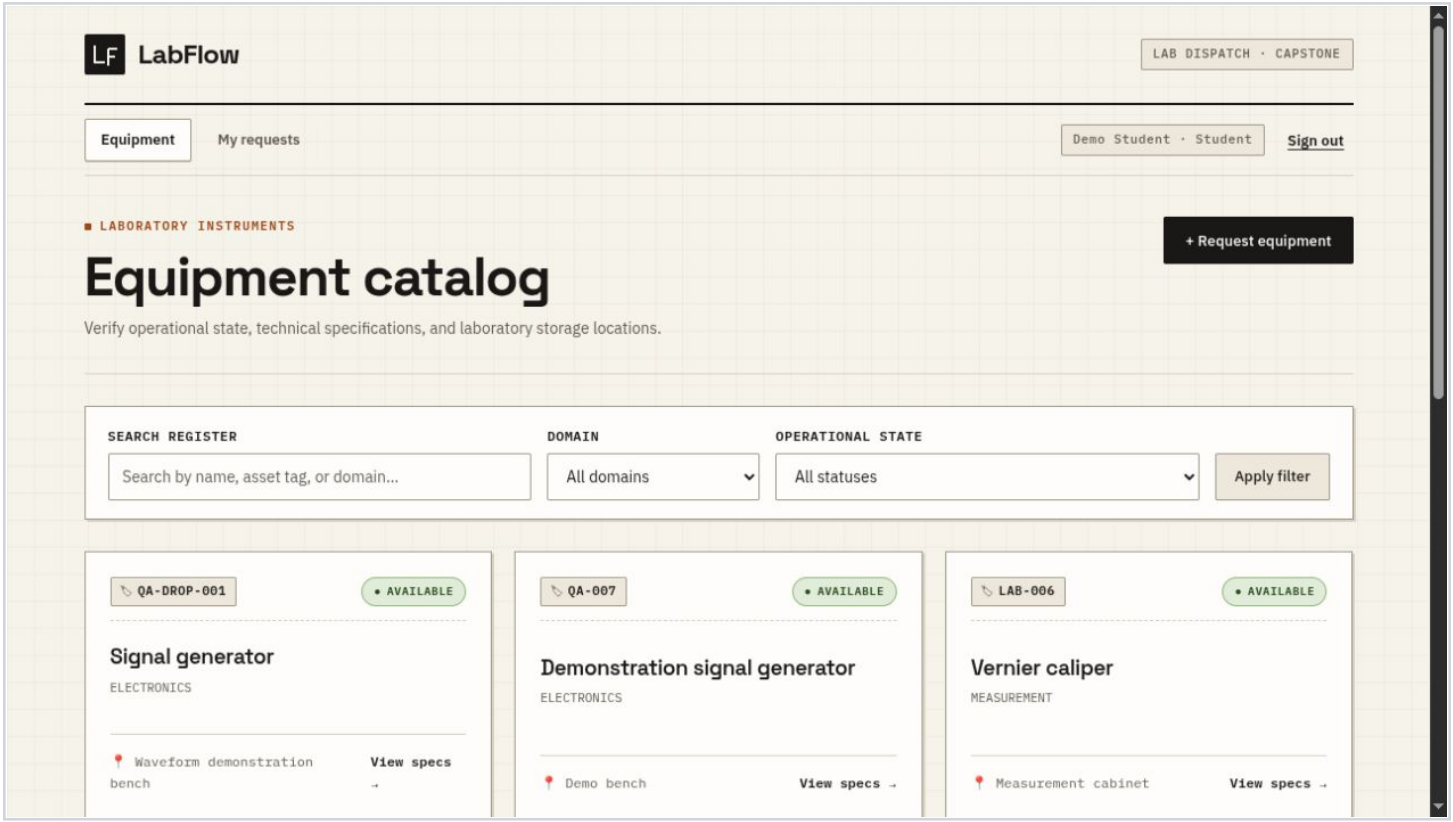
Student registration with strict boundary validation, email normalization, password complexity policy, and atomic duplicate conflict guards.



3. Equipment Catalog & Availability Browser

FEATURE F3 — EQUIPMENT DISCOVERY & SEARCH

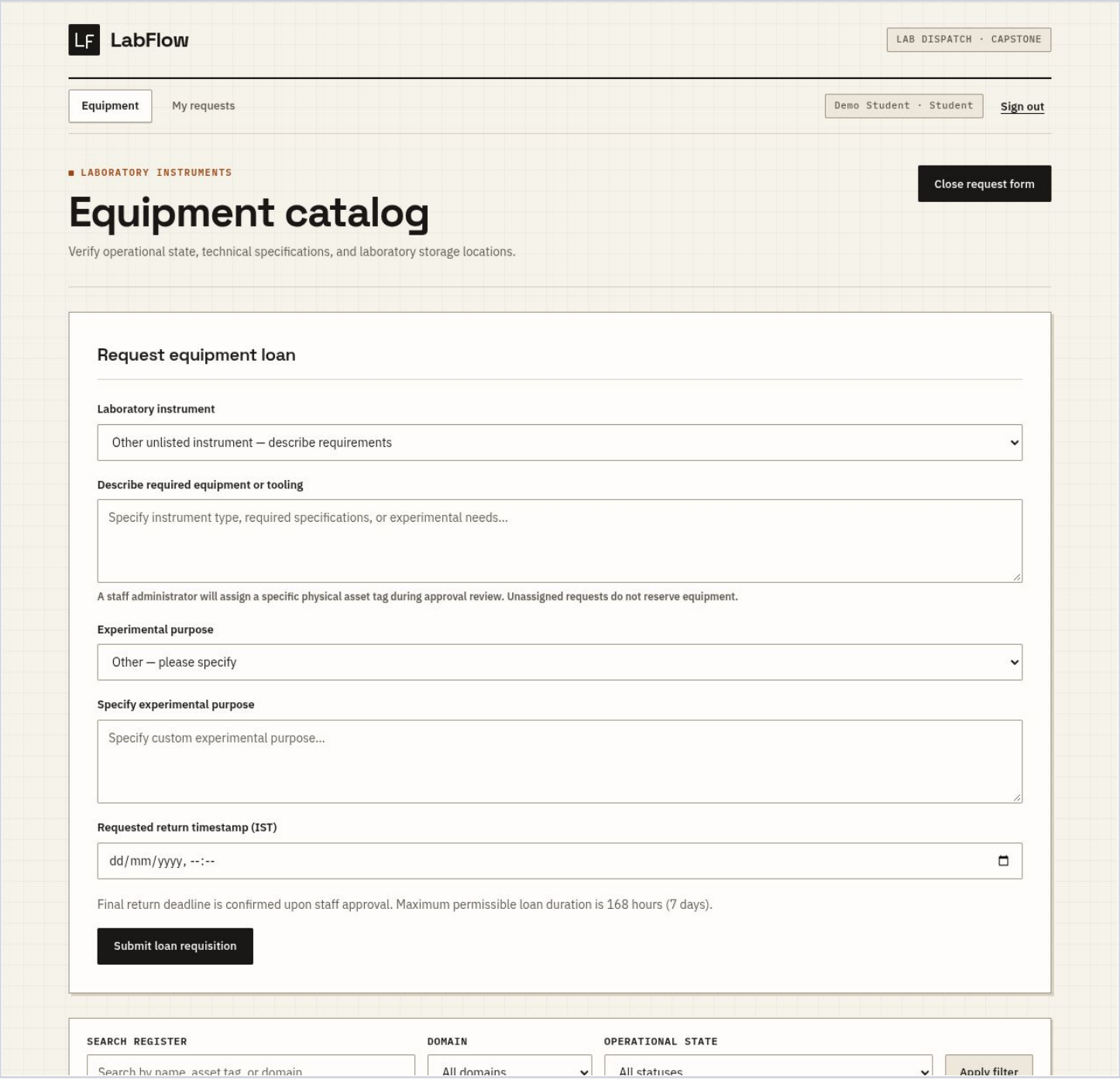
Live searchable inventory catalog displaying individual asset tags, domain classifications, storage locations, and current operational states.



4. Equipment Loan Requisition Form

FEATURE F5 — BORROWING REQUEST & UNLISTED NEEDS

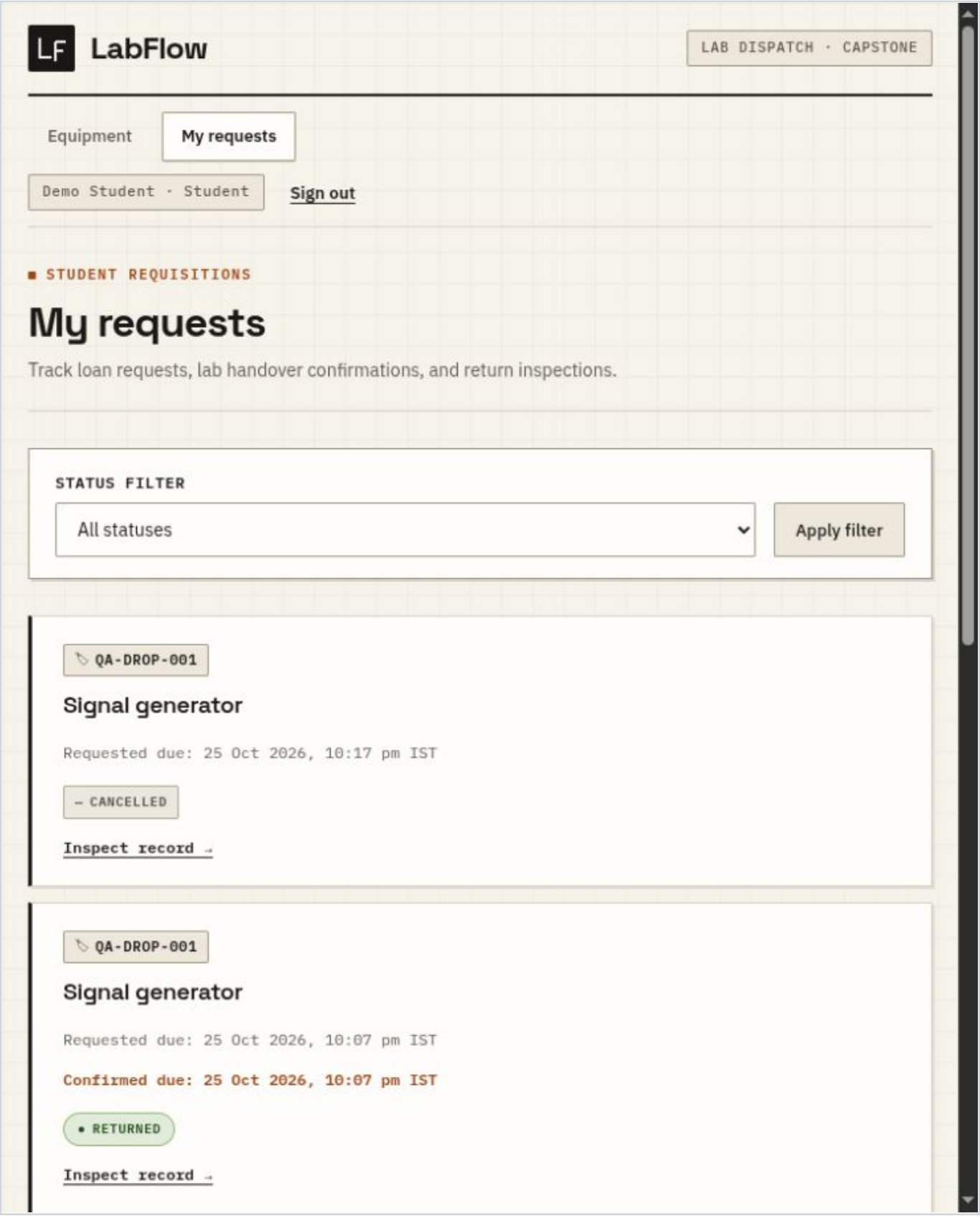
Student requisition form with categorized instrument dropdowns, purpose presets, and free-text description for unlisted custom equipment needs.



5. Student Borrowing History & Status

FEATURE F5 — LOAN LIFECYCLE TRACKING

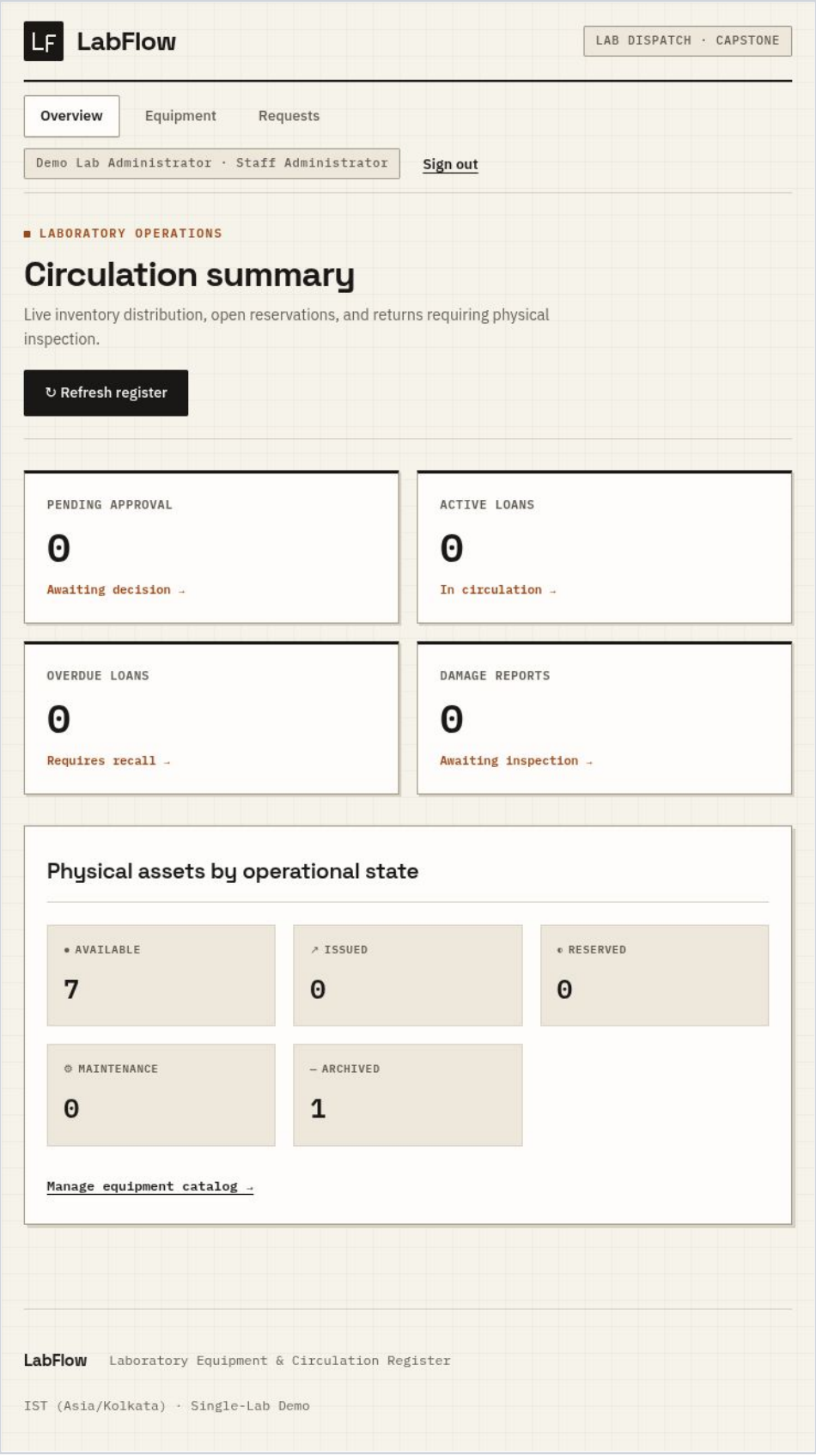
Student dashboard showing personal request timeline, status indicators (Pending, Approved, Issued, Returned), and pre-handover cancellation.



6. Administrator Circulation Dashboard

FEATURE F10 — OPERATIONAL OVERVIEW & SUMMARY

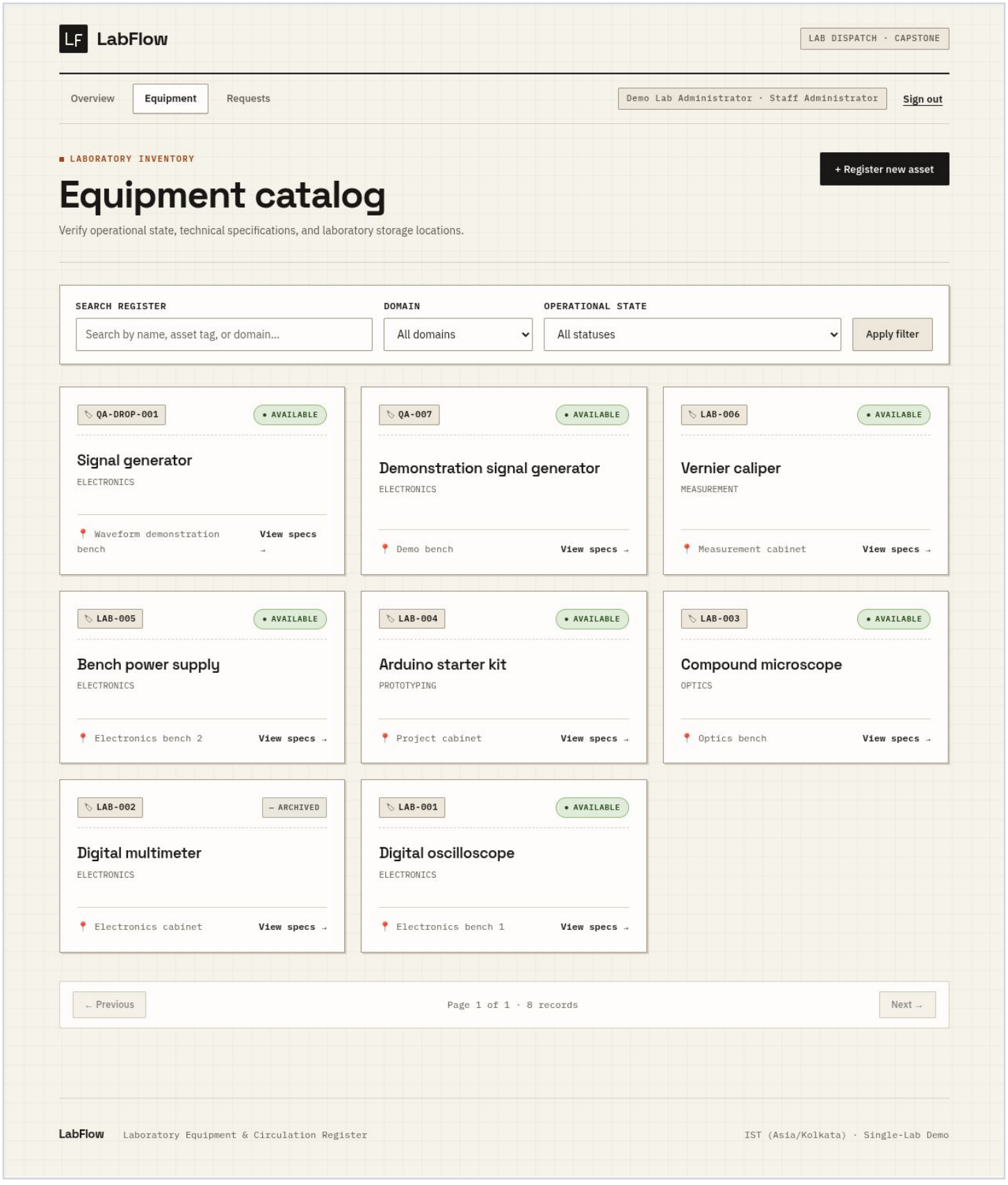
Real-time laboratory overview tracking pending approvals, active circulation counts, overdue loans, and pending damage condition inspections.



7. Laboratory Inventory Register

FEATURE F4 — INVENTORY MANAGEMENT & ASSET TAGGING

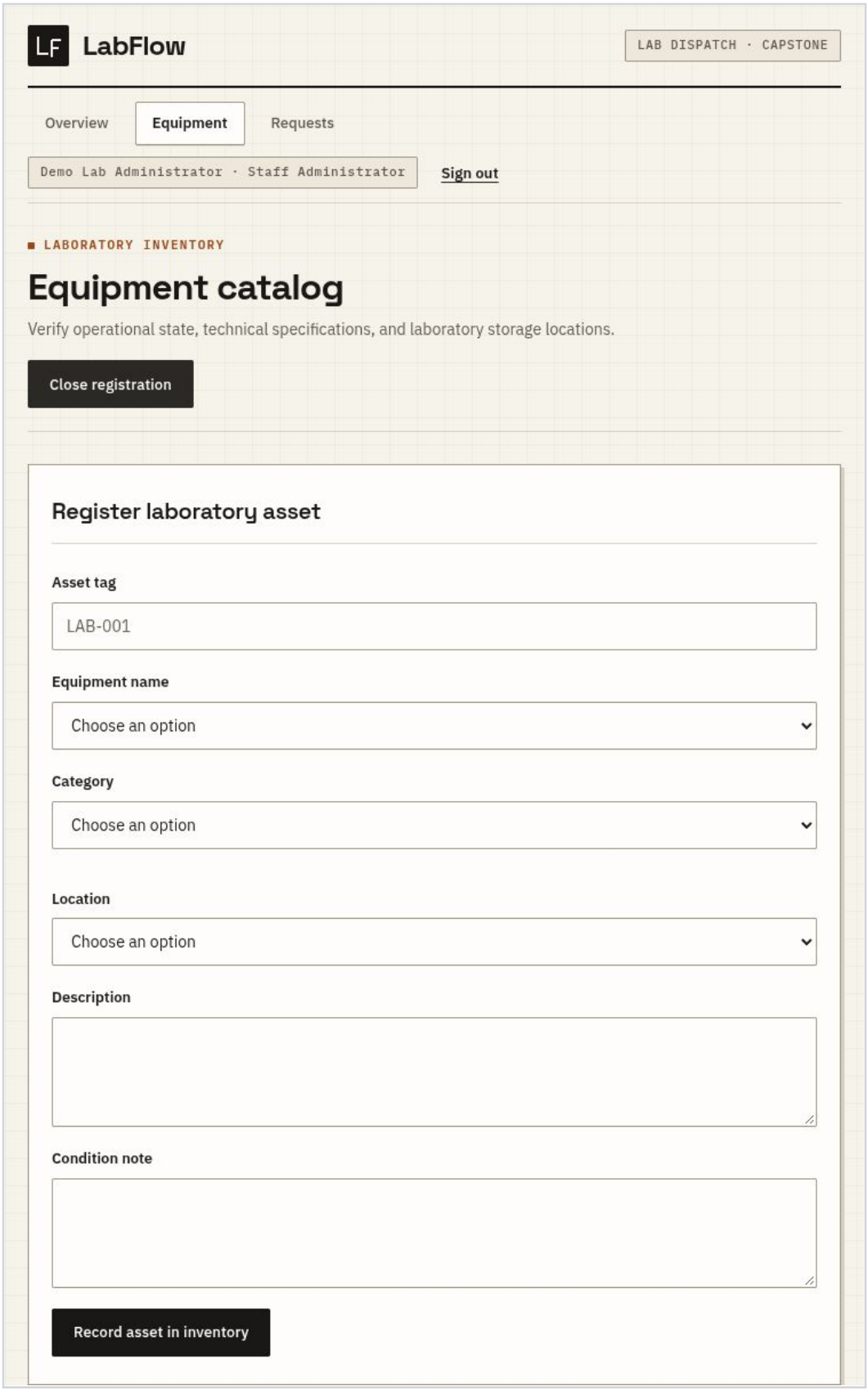
Complete physical asset catalog with operational state filters (Available, Reserved, Issued, Maintenance, Archived) and inline lifecycle actions.



8. Asset Registration & Metadata Entry

FEATURE F4 — ASSET PROVISIONING

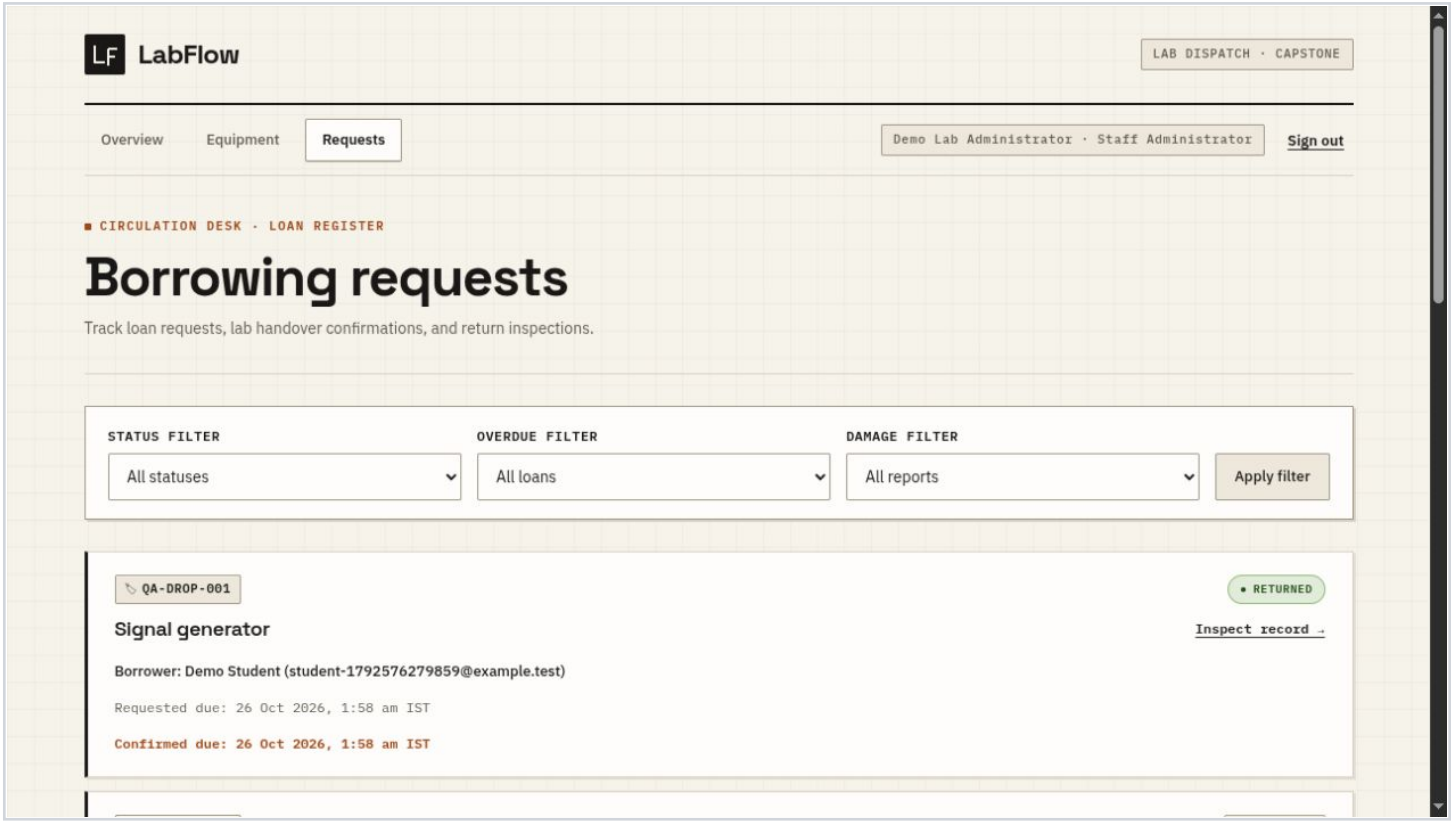
Dialog for registering new physical laboratory instruments with unique asset tags, domain categorization, model specifications, and storage locati



9. Administrator Request Review & Decision

FEATURE F6 & F7 — APPROVAL, ASSIGNMENT & REJECTION

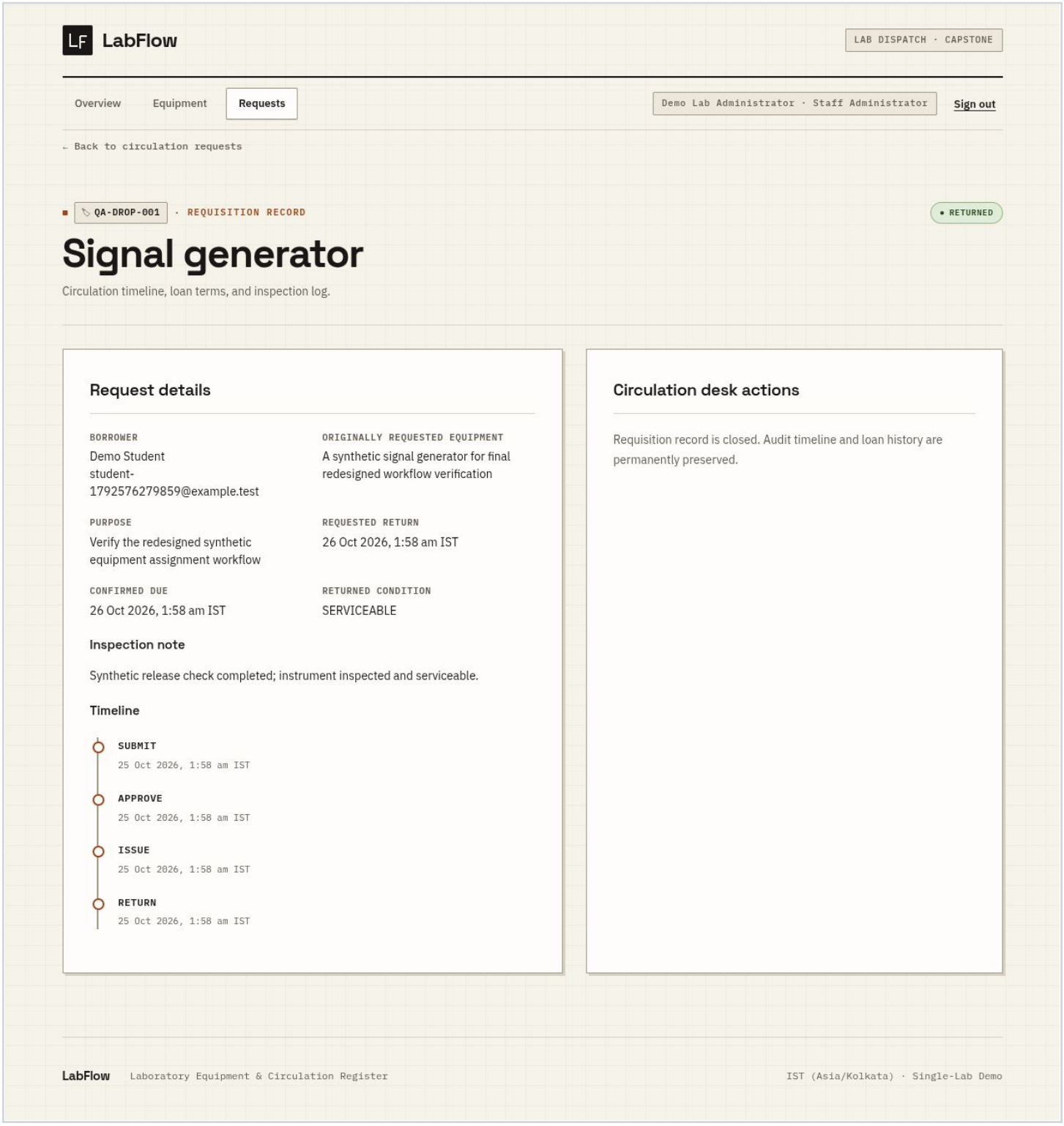
Decision queue allowing staff administrators to review pending student requests, assign available physical assets to unlisted requests, and set loa



10. Inspected Return & Condition Assessment

FEATURE F8 & F9 — EQUIPMENT RETURN & DAMAGE AUDIT

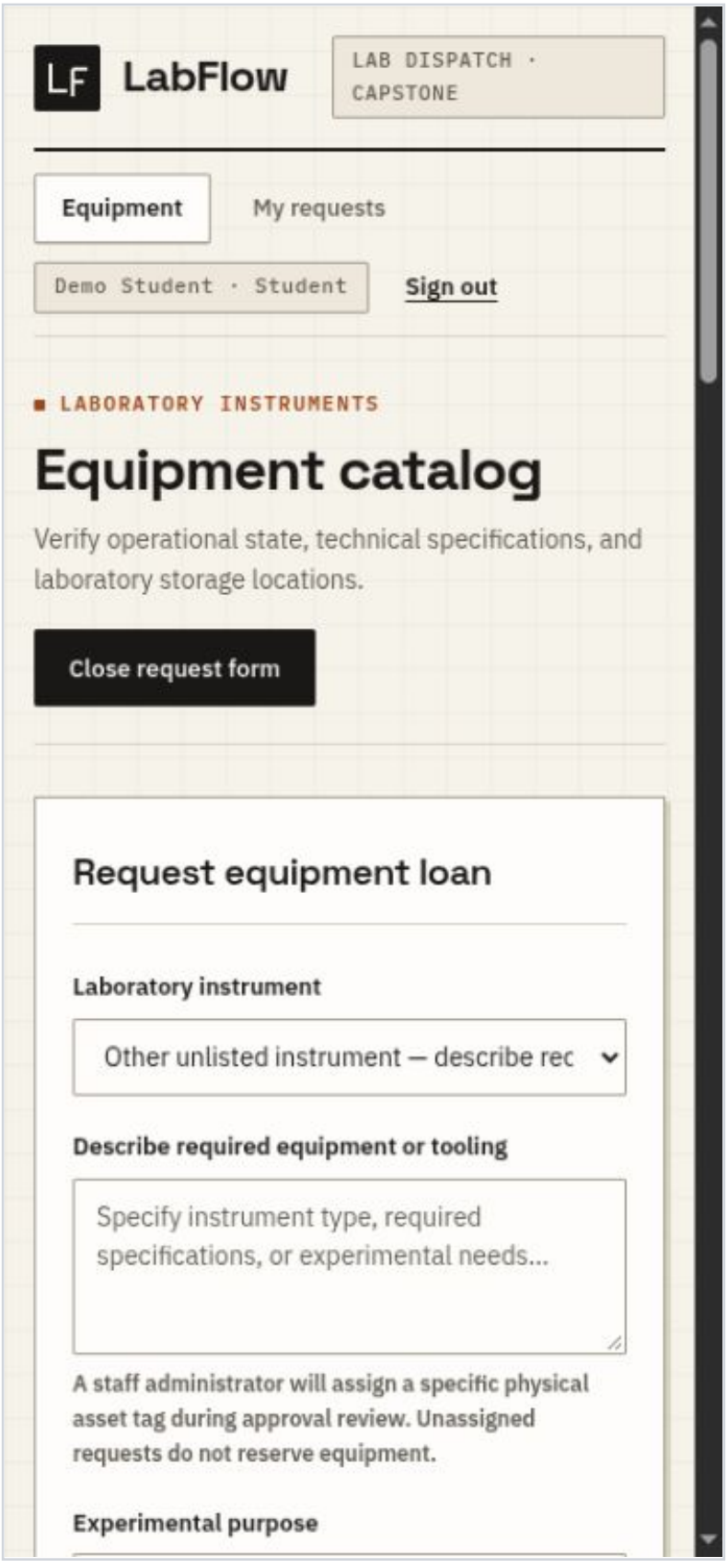
Staff return dialog requiring physical condition verification, recording inspection notes for damaged returns, and auditing complete lifecycle event



11. Mobile Responsive Catalog (320px Viewport)

FEATURE F11 — RESPONSIVE & ACCESSIBLE INTERACTION

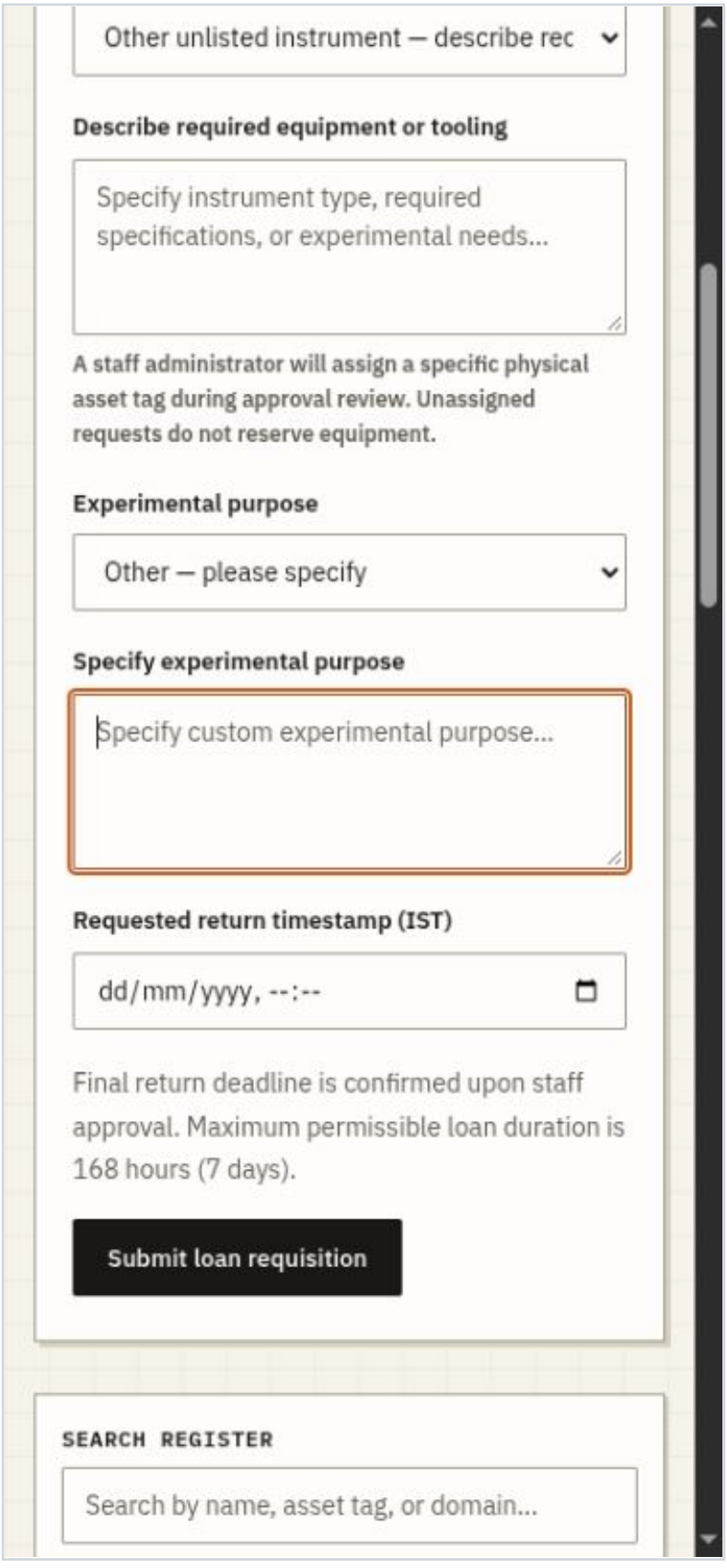
Responsive single-column catalog reflow at 320px viewport without horizontal scrolling, preserving filter controls and touch targets.



12. Mobile Requisition Flow (320px Viewport)

FEATURE F11 — MOBILE FORM CONTROLS

Mobile-optimized borrowing form demonstrating accessible touch inputs, responsive dropdowns, and date picker integration on compact screens.



13. Asset Specification & Operational History

FEATURE F4 & F10 — DETAILED ASSET INSPECTION

Detailed view of an individual laboratory instrument showing serial numbers, maintenance history, current borrower assignment, and audit logs.

